AIVOA PHARMACEUTICALS MANUFACTURING CORP.
GLOBAL QUALITY OPERATIONS & COMPLIANCE DIVISION

PHARMACEUTICAL API QUALITY DEVIATION & INVESTIGATION REPORT

Deviation Number:	DEV-2026-1182	**Classification:**	Major / Polymorphic Crystal Form (CQA)
Product / API Name:	Montelukast Sodium API (Ph. Eur.)	**Batch / Lot Number:**	MTK-2026-215
Manufacturing Stage:	Stage 4: Anti-Solvent Crystallization & Nucleation	**Equipment Tag:**	Crystallizer CRYS-01 / Slurry Pump PMP-102
Date of Occurrence:	2026-10-17 11:20 UTC	**Date of Discovery:**	2026-10-18 15:40 UTC
Material / Lot ID:	Heptane Anhydrous RAW-HEP-901	**Reporting Unit:**	Solid State Characterization Laboratory

1. OBSERVED DEVIATION & ANALYTICAL TEST RESULTS

Test Parameter	Regulatory Specification	Observed Lab Result	Test Methodology / Instrument
Polymorphic Form Purity (XRPD)	**Form I Crystalline NLT 95.0%**	**84.2% Form I (15.8% Amorphous) [OOS]**	X-Ray Powder Diffraction (Bruker D8)
Differential Scanning Calorimetry (DSC)	**Sharp melting peak 135°C - 138°C**	**Broad glass transition at 78°C [OOS]**	TA Instruments DSC 2500
Assay (Anhydrous)	98.0% - 102.0%	99.1%	HPLC 280nm
Specific Optical Rotation	+80.0° to +86.0°	+83.2°	Polarimeter 589nm

2. INCIDENT CHRONOLOGY & FORENSIC NARRATIVE

XRPD solid-state profiling of Montelukast Sodium batch MTK-2026-215 revealed incomplete crystal lattice formation with 15.8% amorphous phase, failing the crystalline specification of NLT 95.0%. Review of SCADA historian records revealed that heptane anti-solvent addition valve FV-102 suffered an automated flow meter calibration error, dumping 45 L/min instead of ramped addition of 15 L/min at 10°C, inducing crash nucleation instead of controlled seed growth.

3. IMMEDIATE CONTAINMENT ACTIONS TAKEN

1. Batch MTK-2026-215 (420 kg) placed on QA hold in conditioned warehouse storage.
2. Flow control loop on CRYS-01 re-calibrated and locked into supervisory control.
3. Recrystallization rework protocol submitted to QA for regulatory risk review.

Prepared by:	Dr. N. Rossi (Solid State Chemist)	2026-10-18
Verified by:	G. Bell (Plant Supervisor)	2026-10-18
QA Approval:	Dr. E. Gomez (QA Director)	**Status: PENDING ROOT CAUSE ANALYSIS**